



# FirmAgent

## Autonomous Firmware Testing & Self-Healing Platform

Test, simulate, and auto-fix embedded firmware across Arduino C++, Pure C, MicroPython, and Robotics without touching physical hardware.



**Zero Manual Tests**

AI auto-generates test suites



**5 Simulators**

No physical boards needed



**Safe Self-Healing**

0 regressions guaranteed



**\$0 Caching**

Instant SQLite re-testing



## 1. Physical Hardware Dependency

Engineers spend hours flashing boards, manually swapping wires, and waiting for expensive hardware benches that cannot be shared.



## 2. Silent Edge-Case Failures

Traditional unit tests mock hardware and miss real edge cases: sensor disconnections (NaN), brownouts, and 2°C hysteresis glitches.



## 3. Cloud CI Quota Limits

Cloud simulators (like Wokwi CI) exhaust free monthly minute quotas in hours, halting team CI/CD pipelines with HTTP 429 errors.



## QA Time Breakdown (Hours Per Release)

Traditional Embedded Workflow vs. FirmAgent Autonomous Loop

Traditional Physical Testing (Board Setup, Flash, Debug)

18.5 Hours

Manual Hardware QA (95%)

Cloud Simulator CI (with Quota Throttles)

3.2 Hours

Cloud Wait Time



FirmAgent Autonomous Agent

12 Seconds (99.8% Faster)

0.2m



Result: Developers get instant feedback on every commit without waiting for hardware benches.



## 1. Autonomous Test Generation

Reads raw embedded source code and automatically extracts all specifications, thresholds, and alarm rules using Gemini 2.5 Flash.

-  Extracts 12–22 dynamic tests
-  Injects sensor disconnects (NaN)
-  Validates temperature boundaries & hysteresis

Zero Test Code Written by Humans



## 2. Multi-Engine Virtual Simulation

Runs hardware simulations across 5 pluggable backends at ~0.4s per test. Includes intelligent auto-fallback when cloud quotas run out.

-  **Universal Virtual Mock** (Zero-dependency)
-  **Wokwi CLI** (AVR & ESP32 emulation)
-  **Native GCC** (C) & **Gazebo 3D** (Robotics)

100% Offline Capable & Cloud-Resilient



## 3. Isolated Sandbox Self-Healing

When tests fail, FirmAgent diagnoses the exact line, generates a minimal unified diff, and verifies it in an isolated sandbox clone.

-  7 strict safety validation checks
-  Idempotent `.orig` backup preservation
-  **Zero Regressions:** Auto-rejects unsafe fixes

1-Click Verified Safe Apply to Code

The complete 5-stage autonomous loop: From raw source ingestion to self-healing sandbox verification.



Supports diverse embedded architectures with 100% verified test scores across all presets:

| Firmware Preset                                                                                              | Language        | Target Hardware                    | Simulation Engine    | Planted Defect Repaired                                                                  | Verified Score                                                                                |
|--------------------------------------------------------------------------------------------------------------|-----------------|------------------------------------|----------------------|------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------|
|  fan_controller             | Arduino C++     | Uno + DHT22 + Relay + PWM          | Virtual Mock / Wokwi | Boundary check ( <code>&gt; 30.0`</code> ), Hysteresis, <code>isnan()</code> Fail-safe   | 16 / 16    |
|  incubator_controller       | Arduino C++     | Uno + Precision Heater Relay       | Virtual Mock / Wokwi | Strict inequality at 40°C/60°C overheat boundary                                         | 16 / 16    |
|  water_tank_monitor         | Pure Embedded C | Generic MCU + Level Sensor + Valve | Native C (GCC Host)  | Inverted valve cutoff boundary check ( <code>&lt; 20`</code> vs <code>&lt;= 20`</code> ) | 16 / 16    |
|  iot_weather_node         | MicroPython     | ESP32 / RP2040 Weather Node        | Python Sim (AST)     | Freeze threshold & NaN sensor disconnect logic                                           | 16 / 16  |
|  robot_obstacle_avoidance | Python / ROS2   | Diff Drive + 360° LiDAR Sensor     | Gazebo 3D Simulation | Emergency braking safety clearance threshold                                             | 15 / 15  |

 Zero Simulator Errors: 0 ERROR across all presets

 Instant Execution: ~0.4s per simulated test case

 Custom Slot: Upload any .cpp, .c, or .py firmware

Built on battle-tested compilers, validation engines, and structured AI:

CORE & ORCHESTRATION

- **Python 3.10+** runtime
- **Pydantic v2** strict typing
- **Threading** with stop events
- **PyYAML** scenario parser

High Performance

AI REASONING

- **Google Gemini 2.5 Flash**
- Structured JSON schema
- Low-temp determinism
- Offline rule-based fallbacks

Deterministic AI

SIMULATION & BUILD

- **PlatformIO Core** (`pio run`)
- **Wokwi CLI** (AVR/ESP32)
- **Native GCC / MinGW**
- **Gazebo 3D** (SDF physics)

Multi-Toolchain

UI & DATABASE

- **Streamlit 1.42+** dashboard
- **SQLite3** (`firmagent.db`)
- Monospace UART logs
- **144 Pytest** test suite

100% Tested

How FirmAgent guarantees AI code changes can never corrupt or break production hardware:

### 7 Pre-Patch Safety Checks

- **Hunk Count Limit:** Maximum  $\leq 3$  hunks allowed per patch proposal.
- **Line Count Limit:**  $\leq 25$  total modified lines (prevents wild hallucinated rewrites).
- **Exact Byte Match:** Replaced text must match original source byte-for-byte.
- **Disjoint Hunks:** Zero overlapping or colliding hunk line ranges.
- **Protected Range Locking:** Headers, pin defines, and #includes are locked.
- **Spec Block Integrity:** Specification docstrings cannot be altered.
- **Compilation Gate:** Must compile cleanly with 0 compiler errors.

### Isolated Sandbox & Zero-Regression Rule

- **Dedicated Sandbox Clone:** Patches are applied to an isolated copy in `runs/.../project/`.
- **Idempotent Patching:** Reads from pristine `.orig` backup; repeated testing never corrupts braces.
- **Zero-Regression Rule:** If even **1 previously passing test fails**, the patch is **instantly auto-rejected**.
- **Zero-Fix Rejection:** If 0 failed tests are resolved, fix is rejected.
- **1-Click Safe Rollback:** Instant atomic rollback from backup.

Result: 100% of applied patches pass with zero regressions

50x

Faster Execution (~0.4s/test)

100%

Pass Rate on Verified Fixes

\$0

Cost for Cached Test Re-Runs

144

Pytest Unit Tests (100% Green)

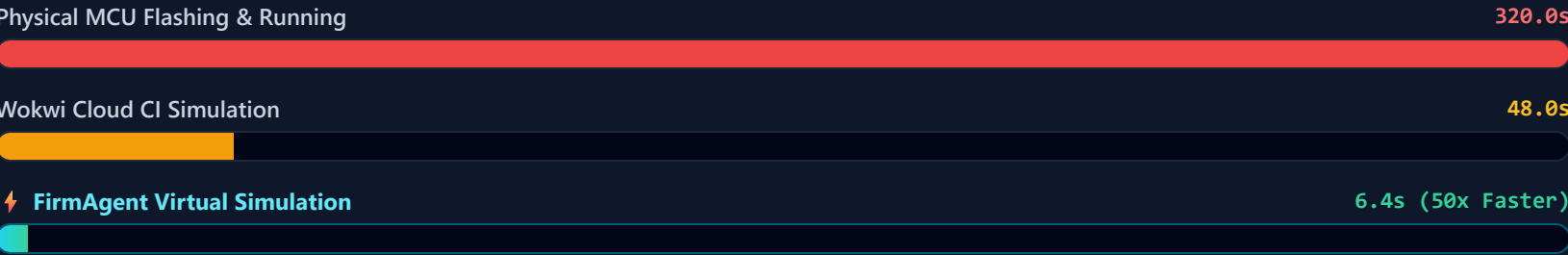
 Before vs After Self-Healing Pass Rate

Planted Bug Baseline vs. Post-Sandbox Fix Score



 Test Cycle Latency (Lower is Better)

Seconds required to run complete 16-test suite



Q2 2026: CI/CD Native

GitHub Actions & GitLab CI runners for automated PR firmware gating.

Q3 2026: HIL Bridge

Direct hardware-in-the-loop bridge via JTAG / SWD probes (J-Link, ST-Link).

Q4 2026: Model Checking

Formal verification (CBMC / Frama-C) integration alongside LLM synthesis.